

Online Experiments for Language Scientists

Lecture 1: Introduction

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What is this course about?

Language is something that humans do, so all subfields of linguistics involve collecting data from humans

- Grammaticality judgments, naturalistic recordings, sociolinguistic interviews, preferential looking/listening in infants, reading times, reaction times and/or choices in psycholinguistic experiments, ...

This kind of data can be collected in person

Or it can be collected online (i.e. via a web browser)

This course shows you how to build language-relevant experiments that run in browsers, and how to crowdsource experiment participants

Who teaches on it?



Kenny Smith

Prof

Course
organiser
Lectures & labs



Alisdair Tullo

*Programming and
Apps Manager*

Labs

How is it delivered?

Lectures (Mondays 10am-10.50am)

Labs (Wednesdays 9am-10.50am)

Lectures

- Each lecture has associated pre-reading, **do the readings before the lectures** so we can discuss any questions/thoughts you have
- Bit of context by me, **plus discussion / Q&A time**
- You can flag up in advance things you'd like to clarify or discuss about the reading in the lecture, via a “quiz” on Learn

Labs: MBA Suite in the Business School* (bring a laptop)

- Work through the practicals, with support on hand!
- Worth at least reading through the materials in advance
- If you don't complete the practicals in lab time, finish them in your own time - try not to fall too far behind
- **Attendance will be taken each week**

* Except in weeks 5-6 when we will be in G.06 50 George Square

Additional drop-in labs

A small number, in the run-in to the final assignment

- Details TBC

For undergraduates, how is it assessed?

Assessment 1: annotated bibliography, worth 30%, due 12th November

- Brief summary plus evaluation of 4 papers
- Papers can come from course readings or elsewhere

Assessment 2: coding project + report, worth 70%, due 10th December

- A functioning web experiment
- A short report explaining the motivation for that experiment, discussing and evaluating critical implementation decisions

Lots of information re. rationale and expectations available in the assignment brief, which will be released this week.

There will be a cut-off date for questions on the assignments!

For **postgraduates**, how is it assessed?

Optional formative assignment: annotated bibliography, worth 0%, due 12th November

- Brief summary plus evaluation of 4 papers
- Papers can come from course readings or elsewhere

Assessment: coding project + report, worth 100%, due 10th December

- A functioning web experiment
- A short report explaining the motivation for that experiment, discussing and evaluating critical implementation decisions

Lots of information re. rationale and expectations available in the assignment brief, which will be released this week.

There will be a cut-off date for questions on the assignments!

Where can I find all this information?

Course Learn page

- Links to course pages on github
- Learn quizzes
- Assignment submission links

Course pages on github

- Everything else

Industrial dispute, strike action

- University management plan to reduce the University's annual budget by £140 million, £90 million from staffing (up to 2000 jobs)
- The main staff union (UCU) believe there is a capital overspending and mismanagement problem, not an over-staffing problem
- UCU want management to commit to no redundancies; management will not
- UCU members have voted for 5 weeks' action starting Monday of week 2 until management agrees to protect our jobs and your education
- **But ... going on strike is expensive, this is the latest in a long-running series of disputes, and recent strikes have not been effective in achieving our goals**
- **I am not currently planning to go on strike**

STUDENT-STAFF ASSEMBLY

FORCED COURSE CLOSURES

£50m of UNETHICAL INVESTMENTS

PROGRAMME CUTS

MASS STAFF CUTS

The university management has undermined and ignored staff and students for too long.

COME TOGETHER, HEAR EACH OTHER, STAND UNITED

24/09/2026 - 17:00-19:00 - APPLETON TOWER, LECTURE THEATRE 4

scan to register

Supported by:

UCU University and College Union

EUPS

unite the UNION

UNISON

Any questions on course organization, admin?

Three components of running an online experiment

Building an experiment that will run in a web browser

- We'll be using javascript and jsPsych
- Also useful for running experiments in-person!

Making it openly available online

- PPLS / the Uni provide servers

Connecting with experiment participants

- E.g. through **crowdsourcing websites**

A look at some simple experiments

Javascript and jsPsych

Javascript: a programming language that runs in web browsers

jsPsych: a library that makes it easy to build experiments

(<https://www.jspsych.org>)

de Leeuw, J. R. (2015). jsPsych: A JavaScript library for creating behavioral experiments in a web browser. *Behavior Research Methods*, 47, 1-12. doi:10.3758/s13428-014-0458-y.



**Josh de
Leeuw**



Plugins and timelines

Plugins: basic building blocks

```
var hello_trial = {  
  type: jsPsychHtmlKeyboardResponse,  
  stimulus: 'Hello world!'  
}
```

Timeline: a sequence of those building blocks

```
jsPsych.run([hello_trial]);
```

A wide range of plugins available

See <https://www.jspsych.org/latest/plugins/list-of-plugins/>

Building an experiment involves

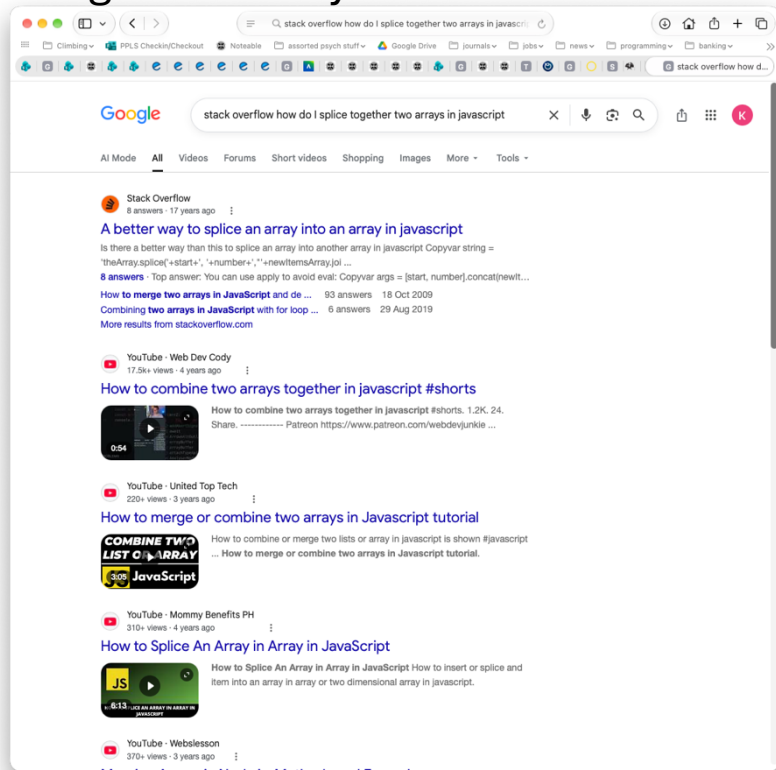
- Knowing how to use plugins
- Figuring out how to piece them together to make the experiment you want
- Some tiny bits of html and javascript to connect the plugins and make them do what you want
- (Occasionally, and optionally, making your own plugin)

A quick word about coding
and realistic expectations!

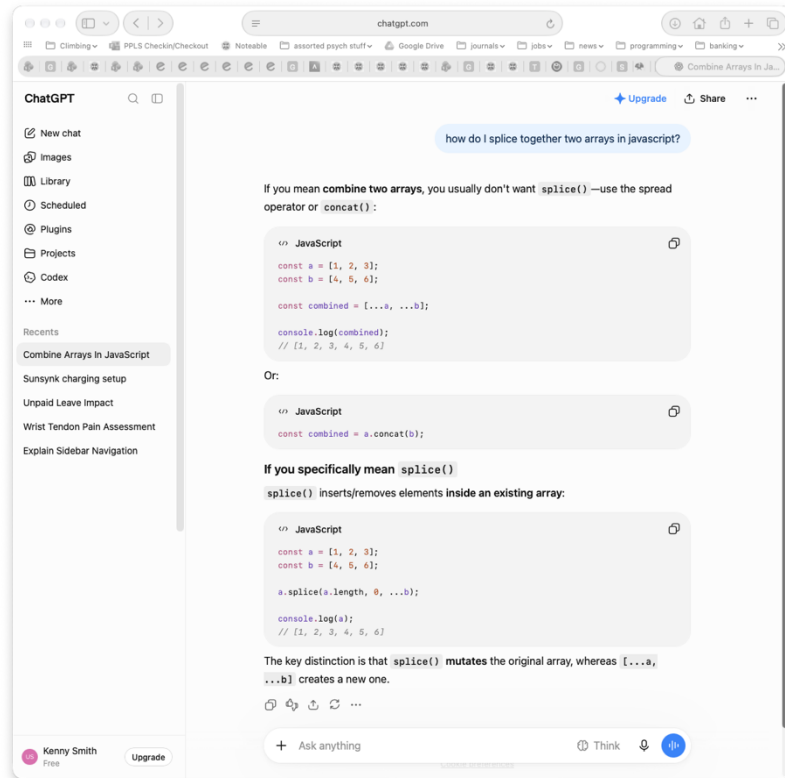


A longer word about Generative AI and coding

The good old days



Now...



A longer word about Generative AI and coding

- Using GenAI to help you figure out how to do small things is OK
- Do not use it to generate more than a line or two of code for you – you will not learn much from this, and code you don't understand is hard to fix / edit!
- **Definitely** do not use it to generate large chunks of code for your final assignment – this puts you at risk of committing academic misconduct
- There is an AI use disclosure form for each assignment

A quick word about reading experimental papers

pared to pronouns. To test whether we can capture this effect using AMT, a linear mixed-effects model was fit using residual reading time as the dependent variable and subject type as a fixed effect. Results indicate that at the subject region, pronouns are read significantly faster than DPs ($\beta = -0.508 \pm 0.02$, $p < 0.0001$). This is illustrated in Fig. 2.

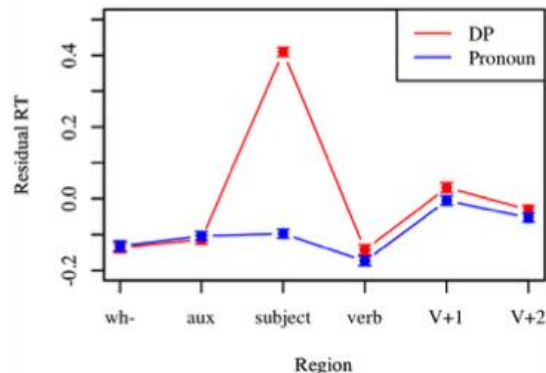


Fig 2. Subject definiteness results. Mean residual reading time is plotted by region for DP and pronoun sentences in Experiment 1. Error bars represent standard error of the mean.

Any questions/concerns so far?

Next up

Wednesday, 9am, MBA Suite, Business School: first lab!

- Week 1 practical, linked from the course page on github
- Bring a laptop
- You'll get more out of the lab if you take a look at the materials beforehand!

Next lecture: crowdsourcing experimental data

- Monday 23rd September
- **Do the reading beforehand!**